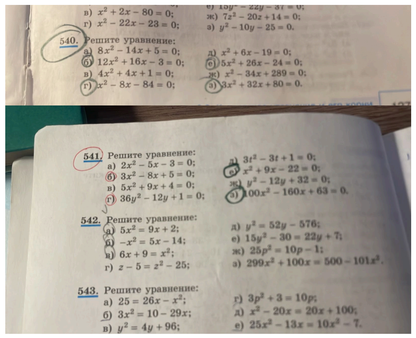


540. Решите уравнение:
а) $8x^2 - 14x + 5 = 0$; д) $x^2 + 6x - 19 = 0$;
б) $12x^2 + 16x - 3 = 0$; е) $5x^2 + 26x - 24 = 0$;
в) $4x^2 + 4x + 1 = 0$; ж) $x^2 - 34x + 289 = 0$;
г) $x^2 - 8x - 84 = 0$; з) $3x^2 + 32x + 80 = 0$.

541. Решите уравнение:
а) $2x^2 - 5x - 3 = 0$; а) $3t^2 - 3t + 1 = 0$;
б) $3x^2 - 8x + 5 = 0$; б) $x^2 + 9x - 22 = 0$;
в) $5x^2 + 9x + 4 = 0$; в) $y^2 - 12y + 32 = 0$;
г) $36y^2 - 12y + 1 = 0$; г) $100x^2 - 160x + 63 = 0$.

542. Решите уравнение:
а) $5x^2 = 9x + 2$; д) $y^2 = 52y - 576$;
б) $-x^2 = 5x - 14$; е) $15y^2 - 30 = 22y + 7$;
в) $6x + 9 = x^2$; ж) $25p^2 = 10p - 1$;
г) $z - 5 = z^2 - 25$; з) $299x^2 + 100x = 500 - 101x^2$.

543. Решите уравнение:
а) $25 = 26x - x^2$; г) $3p^2 + 3 = 10p$;
б) $3x^2 = 10 - 29x$; д) $x^2 - 20x = 20x + 100$;
в) $y^2 = 4y + 96$; е) $25x^2 - 13x = 10x^2 - 7$.

$ax^2 + bx + c = 0$

$D = b^2 - 4ac$

$D > 0 \Rightarrow$ 2 корня

$D = 0 \Rightarrow$ 1 корень

$D < 0 \Rightarrow$ 0 корней

$x_1 = \frac{-b + \sqrt{D}}{2a}$

$x_2 = \frac{-b - \sqrt{D}}{2a}$

$2x^2 - 8 = 0$

$2x^2 = 8$

$x^2 = 4$

$x = 2; -2$

$x^2 + 2x + 1^2 = 0$

$a^2 + 2ab + b^2 = (a+b)^2$

$x = a$

$1 = b$

$(x+1)^2 = 0$

$x + 1 = 0$

$x = -1$

$x^2 + 2x + 1 = 0$

$a = 1$ $D = b^2 - 4ac =$

$b = 2$

$c = 1$ $= 2^2 - 4 \cdot 1 \cdot 1 = 4 - 4 = 0$

1 корень

$x = \frac{-b}{2a} = \frac{-2}{2 \cdot 1} = -\frac{2}{2} = -1$

свернуть в полный квадрат

$a^2 + 2ab + b^2 \longrightarrow (a+b)^2$

$x^2 + 6x + 5 = 0 \Rightarrow x^2 + 6x + 9 - 9 + 5 = 0$

x^2 $2 \cdot a \cdot b$

$a^2 + 2ab + b^2 \to (a+b)^2$

$a = x;\ b = 3$

$(x+3)^2 - 4 = 0$

$(x+3)^2 = 4$

$x + 3 = 2$ $x + 3 = -2$

$x = 2 - 3$ $x = -2 - 3$

$x = -1$ $x = -5$

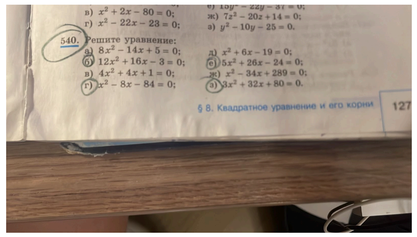
$a = x$

$2ab = 6x$

$2xb = 6x$

$2b = 6;\ b = 3$

$D > 0$: 2к.
$D = 0$: 1к.
$D < 0$: 0к.



$ax^2 + bx + c = 0$

г) $1 \cdot x^2 - 8x - 84 = 0$

1. $a = 1$ 2. $D = b^2 - 4ac = (-8)^2 - 4 \cdot 1 \cdot (-84) =$
$b = -8$
$c = -84$ $= 64 - 4 \cdot (-84) = 64 + 336 = 400 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{8 + \sqrt{400}}{2 \cdot 1} = \frac{28}{2} = 14$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{8 - \sqrt{400}}{2 \cdot 1} = \frac{8 - 20}{2} = \frac{-16}{2} = -8$

Ответ: 14; -8

б) $a(x^2) + bx + c = 0$

$12(x^2) + 16x - 3 = 0$

1. $a = 12$ 2. $D = b^2 - 4ac = 16^2 - 4 \cdot 12 \cdot (-3) =$
$b = 16$
$c = -3$ $= 256 - 48 \cdot (-3) = 256 + 144 = 400 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{-16 + \sqrt{400}}{24} = \frac{-16 + 20}{24} = \frac{1}{6}$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{-16 - 20}{24} = \frac{-36}{24} = -\frac{6}{4} = -\frac{3}{2}$

Ответ: $\frac{1}{6}$; $-\frac{3}{2}$

з) $3x^2 + 32x + 80 = 0$

1. $a = 3$ 2. $D = b^2 - 4ac = 32^2 - 4 \cdot 3 \cdot 80 =$
$b = 32$
$c = 80$ $= 1024 - 960 = 64 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{-32 + \sqrt{64}}{6} = \frac{-32 + 8}{6} = -\frac{24}{6} = -4$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{-32 - 8}{6} = -\frac{40}{6} = -\frac{20}{3}$

Ответ: -4; $-\frac{20}{3}$

$ax^2 + bx + c = 0 \quad / : a$

$\frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} = 0$

$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \Rightarrow x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$

x^2 $2yz$

$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$

$y = x$

$2yz = \frac{b}{a}y$

$2z = \frac{b}{a} \Rightarrow z = \frac{b}{2a}$

$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$ (4a)

$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$

$y = x + \frac{b}{2a}$ $y^2 = z$

$z = \frac{b^2 - 4ac}{4a^2}$ $y = \sqrt{z}$ $y = -\sqrt{z}$

Обр. замена:

$x + \frac{b}{2a} = \sqrt{\frac{b^2 - 4ac}{4a^2}}$ $x + \frac{b}{2a} = -\sqrt{\frac{b^2 - 4ac}{4a^2}}$

$\sqrt{9-4} = \sqrt{9} - \sqrt{4}$ ✗

$\sqrt{9 \cdot 4} = \sqrt{9} \cdot \sqrt{4}$ ✓

$x = -\frac{b}{2a} + \sqrt{\frac{b^2 - 4ac}{4a^2}}$ $x = -\frac{b}{2a} - \sqrt{\frac{b^2 - 4ac}{4a^2}}$

$x = -\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$ $x = -\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$

$x = -\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a}$ $x = -\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

$b^2 - 4ac = D$ - дискриминант.

е) $5x^2 + 26x - 24 = 0$

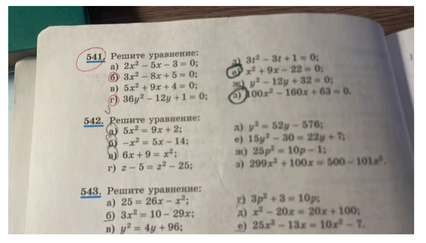
1. $a = 5$ 2. $D = b^2 - 4ac = 26^2 - 4 \cdot 5 \cdot (-24) =$
$b = 26$
$c = -24$ $= 676 + 480 = 1156 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{-26 + 34}{10} = \frac{8}{10} = 0{,}8$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{-26 - 34}{10} = \frac{-60}{10} = -6$

Ответ: 0,8; -6



б) $3x^2 - 8x + 5 = 0$

1. $a = 3$ 2. $D = b^2 - 4ac = 64 - 4 \cdot 3 \cdot 5 =$
$b = -8$
$c = 5$ $= 64 - 60 = 4 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{8 + 2}{6} = \frac{10}{6} = \frac{5}{3}$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{8 - 2}{6} = \frac{6}{6} = 1$

Ответ: $\frac{5}{3}$; 1

г) $36y^2 - 12y + 1 = 0$

1. $a = 36$ 2. $D = b^2 - 4ac = 144 - 4 \cdot 36 \cdot 1 =$
$b = -12$
$c = 1$ $= 144 - 144 = 0$

3. 1 корень

$x = \frac{-b}{2a} = \frac{12}{2 \cdot 36} = \frac{1}{6}$

Ответ: $\frac{1}{6}$

е) $x^2 + 9x - 22 = 0$

1. $a = 1$ 2. $D = b^2 - 4ac = 81 - 4 \cdot 1 \cdot (-22) =$
$b = 9$
$c = -22$ $= 81 + 88 = 169 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{-9 + \sqrt{169}}{2 \cdot 1} = \frac{-9 + 13}{2} = \frac{4}{2} = 2$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{-9 - 13}{2} = \frac{-21}{2}$

Ответ: 2; $-\frac{21}{2}$

ж) $100x^2 - 160x + 63 = 0$

1. $a = 100$ 2. $D = b^2 - 4ac = 25600 - 4 \cdot 100 \cdot 63 =$
$b = -160$
$c = 63$ $= 25600 - 25200 = 400 > 0$

3. 2 корня

$x_1 = \frac{-b + \sqrt{D}}{2a} = \frac{160 + 20}{200} = \frac{180}{200} = \frac{18}{20} = \frac{9}{10}$

$x_2 = \frac{-b - \sqrt{D}}{2a} = \frac{160 - 20}{200} = \frac{140}{200} = \frac{70}{100} = \frac{7}{10}$

Ответ: 0,9; 0,7